

InsureTech

**INSURANCE MANAGEMENT
SYSTEM**

Salesforce-Based Policy and Claim Management Platform

PROJECT REPORT

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Abstract

Insurance Management System is a Salesforce-based application designed to centralize and automate customer, policy, and claim management. The system manages customer information, insurance policies, premium details, claim requests, approvals, reports, and dashboards. Salesforce Flows, Apex Triggers, Validation Rules, Formula Fields, Reports, and Dashboards automate the insurance lifecycle while improving efficiency and data consistency.

1. Introduction

Insurance companies handle large volumes of customer, policy, and claim information. Manual record management often causes duplicate records, delayed claim processing, and inaccurate policy tracking. This project provides a centralized Salesforce solution for managing the complete insurance lifecycle.

2. Problem Statement

Many insurance organizations rely on spreadsheets and manual processes for maintaining customer records, policies, and claims. This leads to slow processing, data inconsistency, and limited business insights. The proposed system automates these activities using Salesforce.

3. Existing System

Customer details, policies, and claims are maintained separately, requiring manual verification and updates. Reporting is also largely manual.

3.1 Limitations of the Existing System

- Customer information stored in multiple places.
- Manual policy number generation.
- Slow claim processing.
- Delayed policy updates.
- Duplicate records.
- Limited analytics.
- Increased human error.

4. Proposed System

The proposed Insurance Management System manages Customers, Policies, and Claims using Salesforce custom objects and lookup relationships. Apex automatically generates policy numbers, Flows activate policies and update policy status after approved claims, while reports and dashboards provide business insights.

4.1 Advantages

- Centralized insurance data.
- Automated policy creation.
- Faster claim processing.
- Improved data accuracy.
- Real-time reports and dashboards.
- Reduced manual effort.

5. Objectives

- Manage customers, policies, and claims.
- Automate policy number generation.
- Validate premium amount.
- Automate policy status updates.
- Generate reports and dashboards.

6. Functional and Non-Functional Requirements

Functional Requirements:

- Manage Customers.
- Create Policies.
- Process Claims.
- Generate Policy Numbers.
- Reports and Dashboards.

Non-Functional Requirements:

- User-friendly Lightning interface.
- Secure data.
- Scalable automation.
- Reliable validation.

7. Technology Stack

Salesforce CRM – Core Platform

Object Manager – Custom Objects and Relationships

Flow Builder – Automation

Apex – Policy Number Generation
Developer Console – Apex Development
Lightning App Builder – UI
Reports & Dashboards – Analytics

8. Phase 1 – Requirement Analysis and Planning

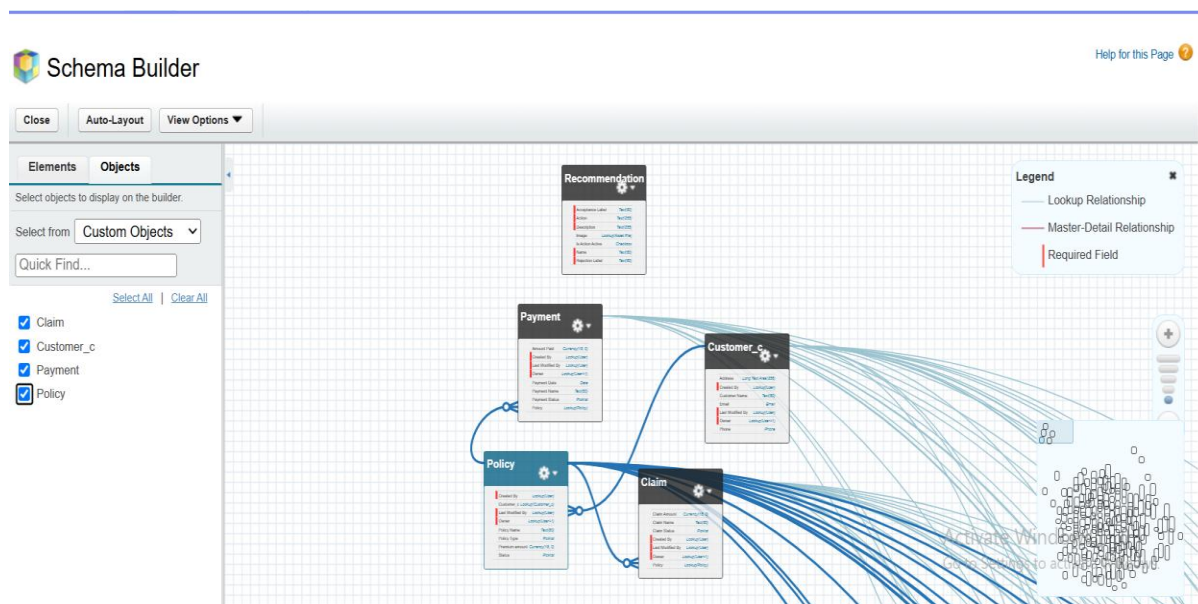
The system requirements identified Insurance Agents, Claim Officers, Policy Managers, and Administrators as major users. Business processes were analyzed before configuring the Salesforce application.

8.1 Data Model

The data model consists of three custom objects: Customer, Policy, and Claim. Customer and Policy are connected using a Lookup relationship, while Policy and Claim are connected through another Lookup relationship.

Customer
↓
Policy
↓
Claim

Figure: Schema Builder and Data Model



9. Phase 2 – Salesforce Development and Configuration

9.1 Customer Management

The Customer object stores customer information such as Customer Name, Phone Number, and Email Address. It serves as the primary record for maintaining customer details and is linked to Policy records using a Lookup Relationship. This centralized approach eliminates duplicate customer records and simplifies policy management.

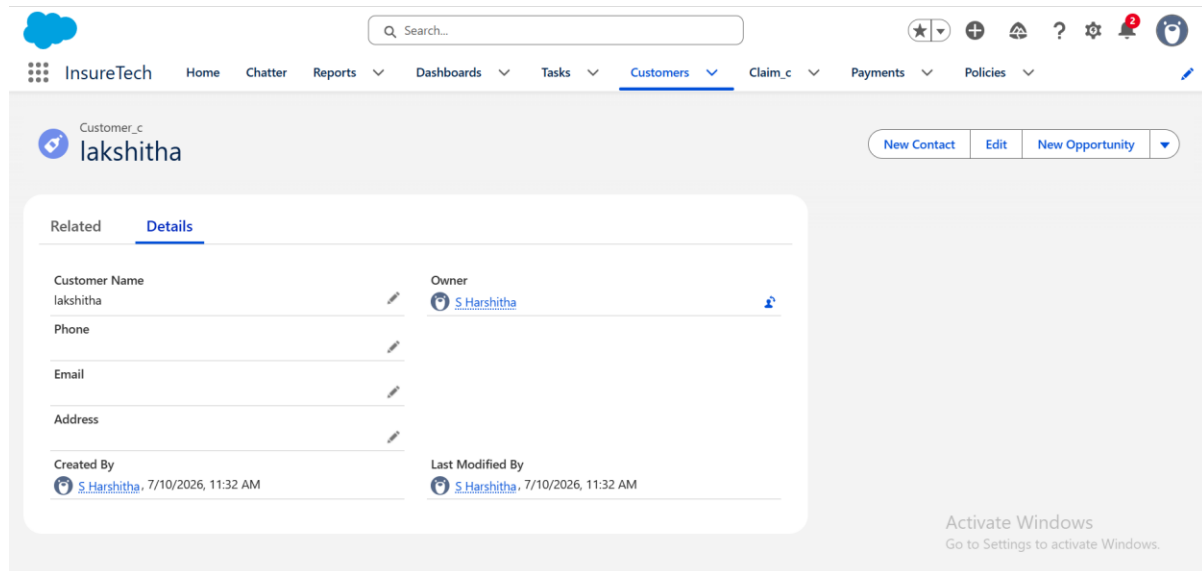


Figure: Customer Record

9.2 Validation Rules

Validation Rules were configured to improve data quality. A validation rule on the Policy object ensures that the Premium Amount must always be greater than zero. Invalid records are prevented from being saved, improving data consistency.

Validation Rule:

Premium_Amount _ c <= 0

Error Message: Premium Amount must be greater than zero.

Policy PAY001

Policy Type: Health

Premium amount: \$0

Premium Amount must be greater than 0.

Status: Active

Customer_c: lakshitha

Created By: S Harshitha, 7/10/2026, 11:46 AM

Updated By: S Harshitha, 7/10/2026, 11:46 AM

We hit a snag. Review the following fields: Premium amount

Cancel Save

Activate Windows Go to Settings to activate Windows.

Figure: Premium Amount Validation Rule

9.3 Formula Field

A Formula Field named 'Policy Active' was created on the Policy object to display whether a policy is currently active. The formula automatically returns 'Yes' when the Policy Status is Active; otherwise, it returns 'No'.

Formula:

IF(Status__c='Active','Yes','No')

InsureTech

Search...

Policy PAY001

Related Details

Policy Name	PAY001	Owner	S Harshitha
Policy Type	Health		
Premium amount	\$100,000		
Status	Active		
Customer_c	lakshitha		
Created By	S Harshitha, 7/10/2026, 11:46 AM	Last Modified By	S Harshitha, 7/11/2026, 9:18 PM

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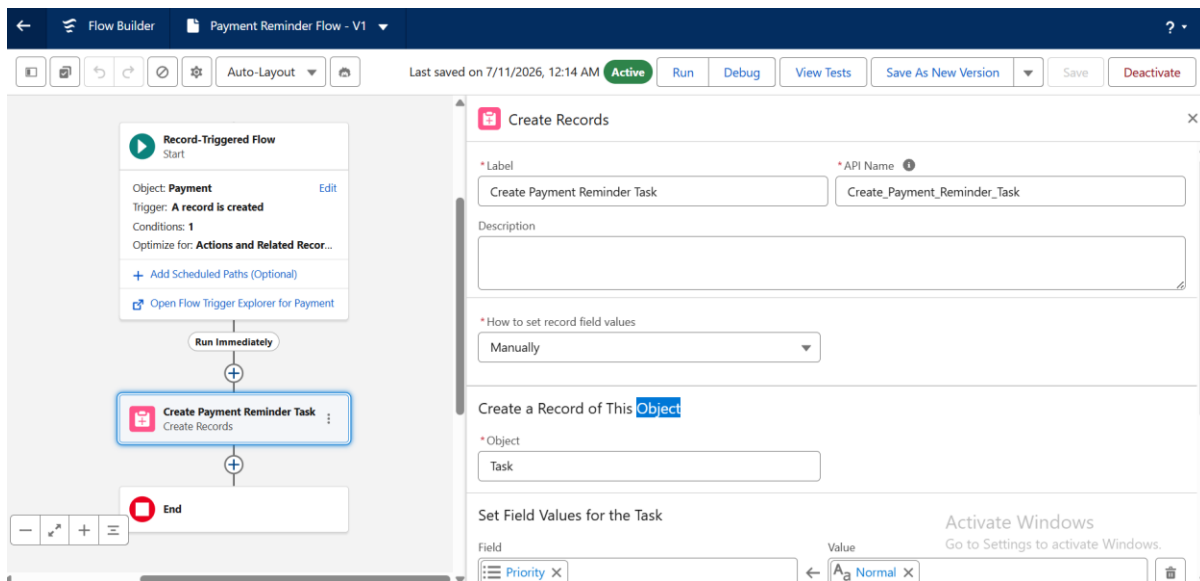
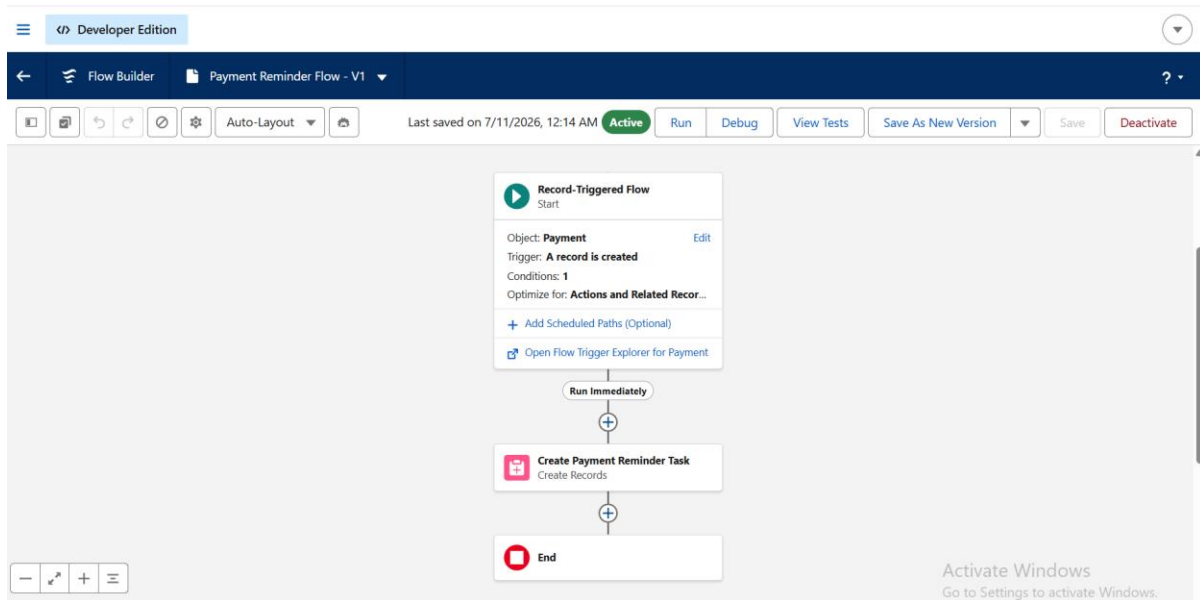
Figure: Formula Field

9.4 Record-Triggered Flow

A Record-Triggered Flow automatically updates business records. When a new Policy is created and its Status is blank, the Flow sets the Status to 'Active'. Another Flow monitors Claim records and updates the related Policy Status to 'Claim Processed' whenever a

Claim is approved.

Figure: Record-Triggered Flow



9.5 Apex Automation

An Apex Trigger was implemented on the Policy object to automatically generate a unique Policy Number during record creation. The trigger assigns values such as POL-1001, POL-1002, and POL-1003, eliminating manual entry and ensuring uniqueness. Business logic can be maintained in a separate handler class for better scalability.

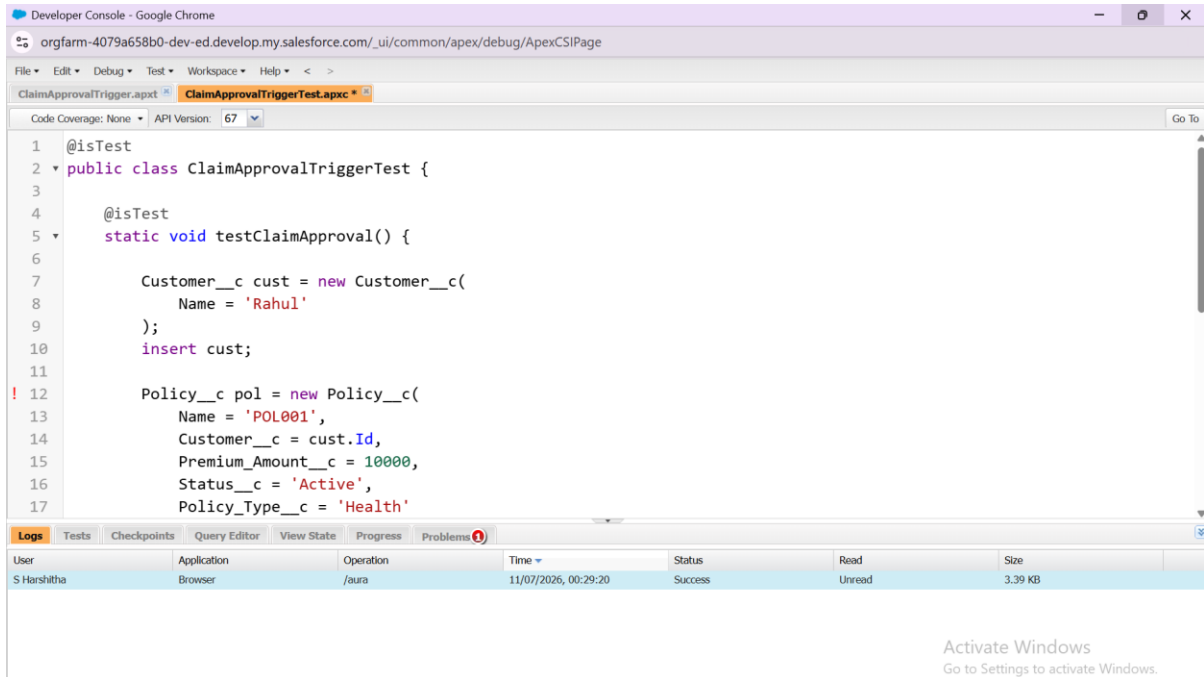


Figure: Apex Trigger Implementation

10. Phase 3 – UI/UX Development and Customization

The Insurance Management System Lightning App provides navigation to Customers, Policies, Claims, Reports, and Dashboards through an intuitive interface. The application allows insurance staff to manage records efficiently from a single workspace.

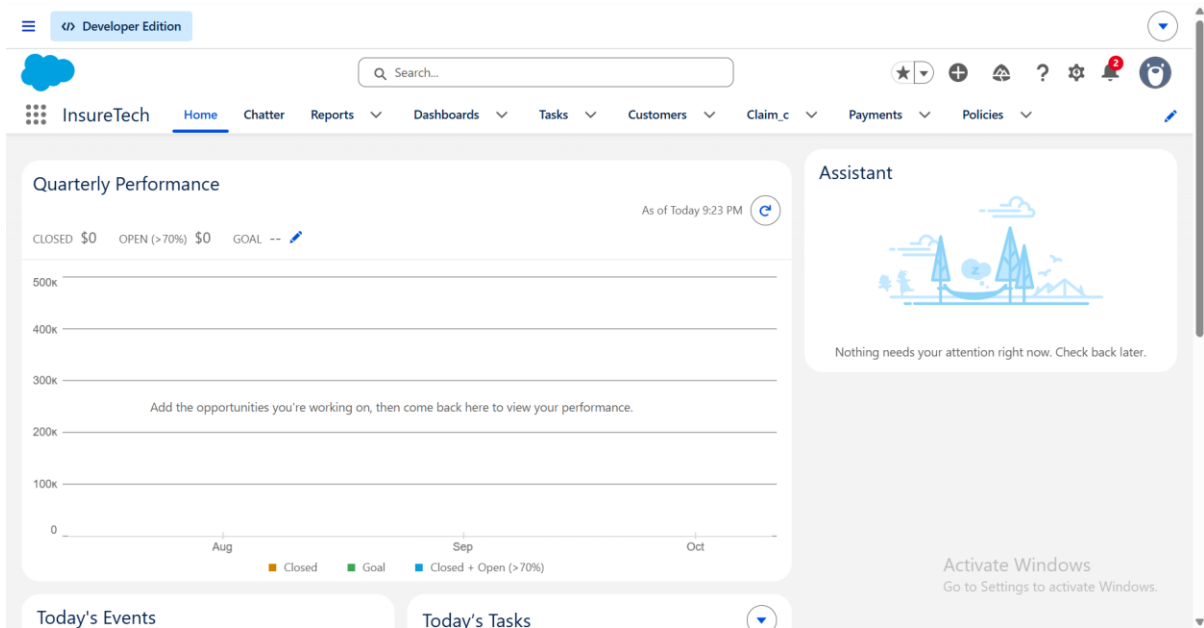


Figure: Insurance Management System Lightning App

10.1 Reports

Reports were created for Customers, Active Policies, Policies by Type, Claims by Status, and Customer Policy Details. These reports summarize operational data and support business decision-making.

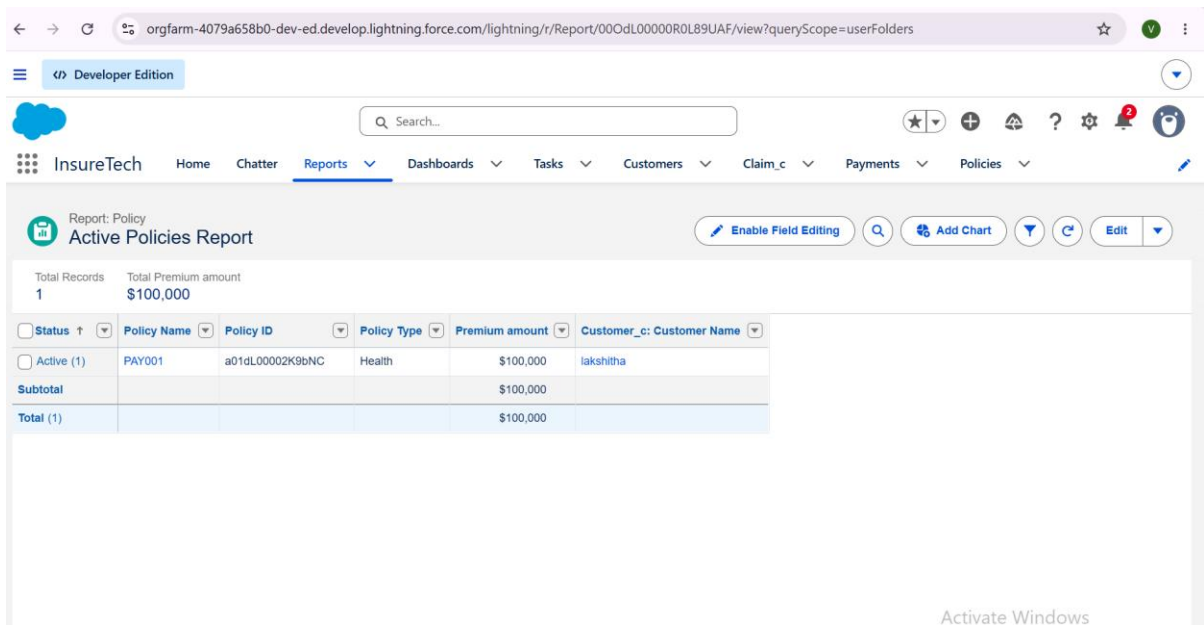


Figure: Salesforce Reports

10.2 Dashboard

The Salesforce Dashboard provides a centralized view of key insurance business metrics using graphical components and real-time analytics. It enables insurance managers and administrators to monitor customer records, policy information, and claim activities from a single interface.

The dashboard includes KPI metrics, bar charts, pie charts, and donut charts that summarize important operational data. By presenting business information visually, the dashboard helps management quickly identify trends, monitor policy performance, analyze claim status, and make informed decisions.

The dashboard automatically refreshes as new records are created or existing records are updated, ensuring that users always have access to the latest information.

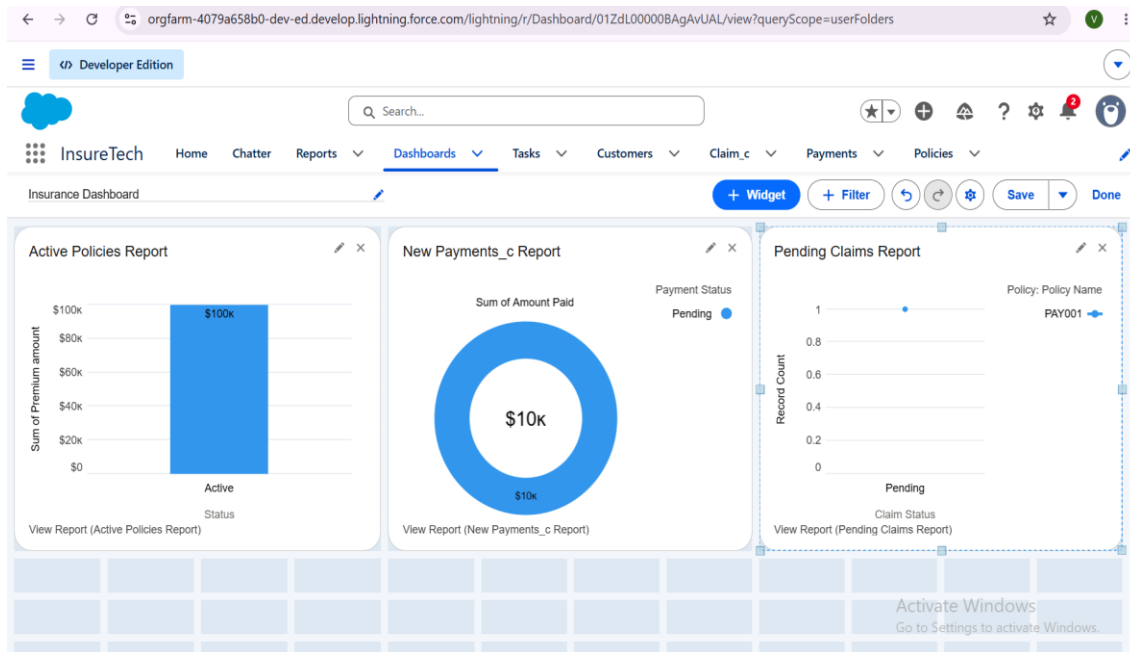


Figure: Insurance Management System Dashboard

11. Detailed System Workflows

11.1 Overall End-to-End Workflow

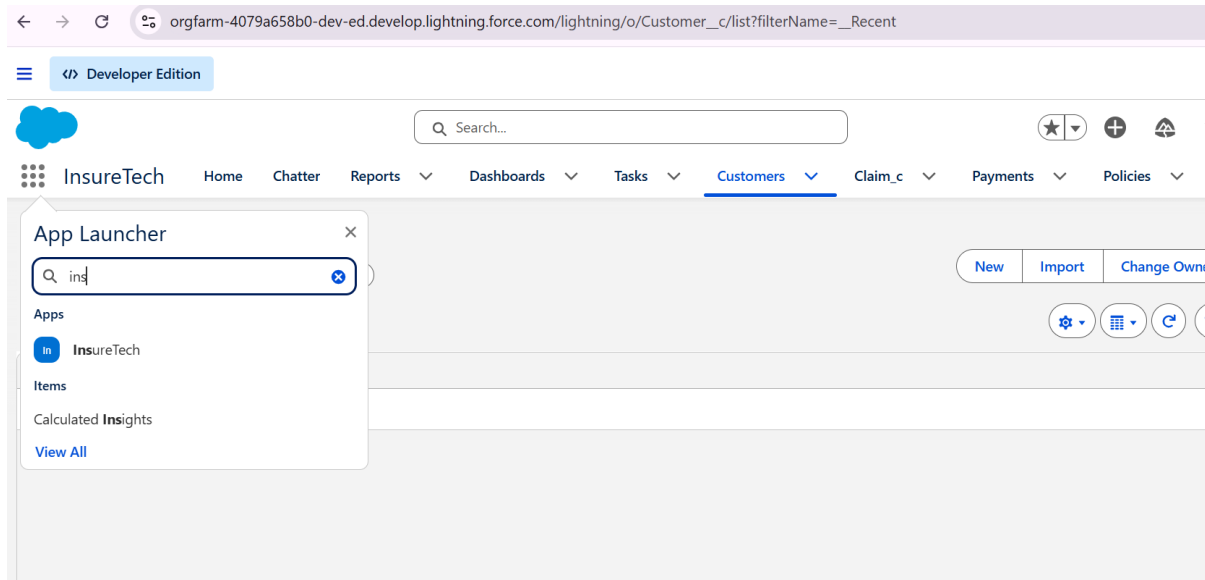
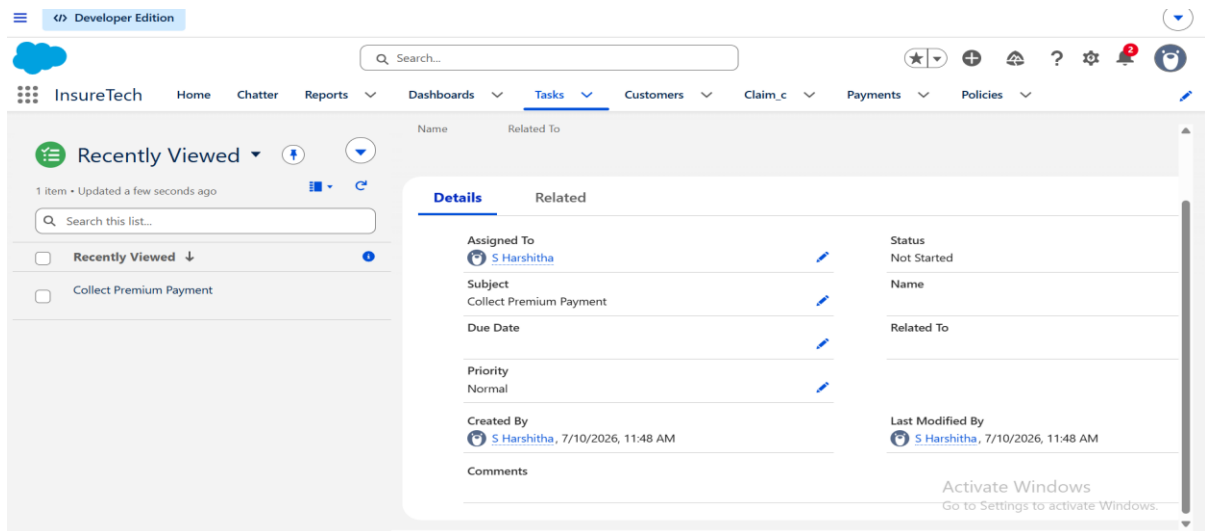
The Insurance Management System follows a complete business workflow beginning with customer registration and ending with business reporting.

Initially, an insurance agent registers a new customer by entering personal details. After customer registration, an insurance policy is created and associated with the customer through a Lookup Relationship.

When the policy record is saved, the Apex Trigger automatically generates a unique Policy Number. If the policy status is left blank, the Record-Triggered Flow automatically updates the status to Active.

If the customer later submits an insurance claim, a Claim record is created and linked to the corresponding Policy. Once the claim is approved, another Record-Triggered Flow automatically updates the Policy Status to Claim Processed.

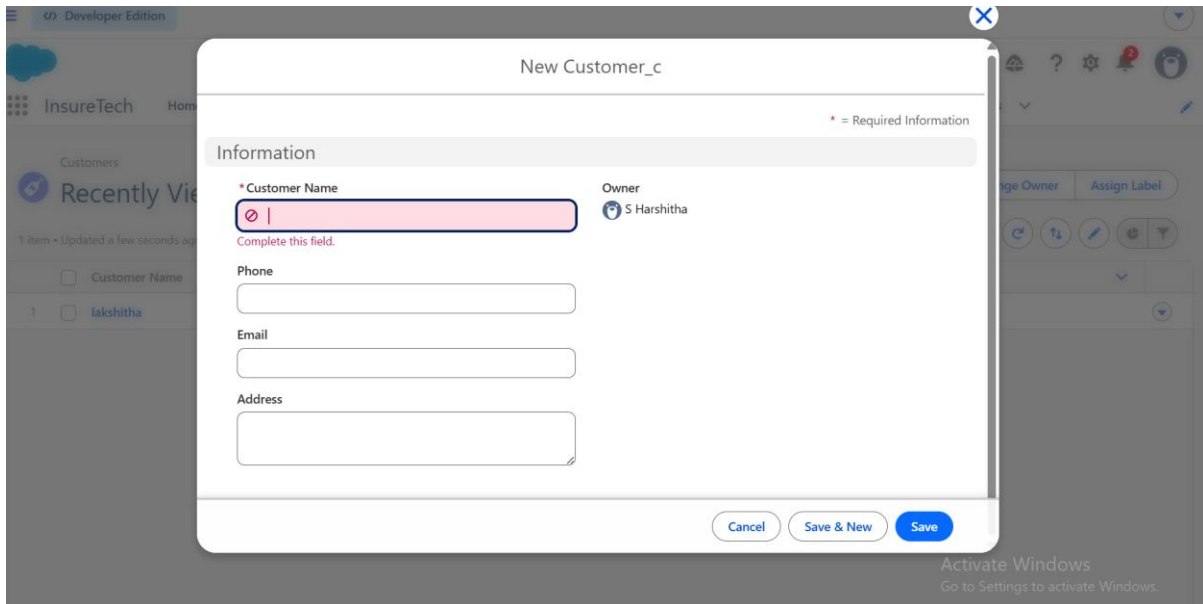
Finally, customer, policy, and claim information is summarized through Salesforce Reports and Dashboards.



11.2 Customer Management Workflow

The Customer Management Workflow begins with customer registration. Insurance agents enter customer details including name, phone number, and email address.

Each customer record is connected with one or more policies using a Lookup Relationship. This workflow reduces duplicate customer records and enables centralized customer management.

A screenshot of a Salesforce interface showing a 'New Customer_c' form. The form is titled 'New Customer_c' and has a close button (X) in the top right corner. It features a 'Required Information' asterisk icon. The form fields include: 'Customer Name' (with a red error message 'Complete this field.'), 'Phone', 'Email', and 'Address'. The 'Owner' field is set to 'S Harshitha'. At the bottom, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The background shows a sidebar with 'InsureTech' and 'Customers' sections, and a 'Recently Viewed' list with one item 'lakshitha'.

11.3 Policy Management Workflow

The Policy Management Workflow begins after customer registration. Insurance agents create a policy by selecting a customer and entering policy details.

The Apex Trigger automatically generates a Policy Number while the Record-Triggered Flow sets the Status to Active when left blank. Validation Rules ensure the Premium Amount is greater than zero.

11.4 Claim Management Workflow

The Claim Management Workflow begins when a customer submits a claim.

A Claim record is linked to its Policy. Initially the claim remains Pending. After review, it may become Approved or Rejected. When approved, the related Policy Status is automatically updated to Claim Processed by Flow.

Policy Automation Workflow

Salesforce automation minimizes manual work. Apex generates Policy Numbers while Record-Triggered Flows activate policies and update policy status after approved claims. This improves efficiency and data consistency.

11.5 Apex Policy Number Generation Workflow

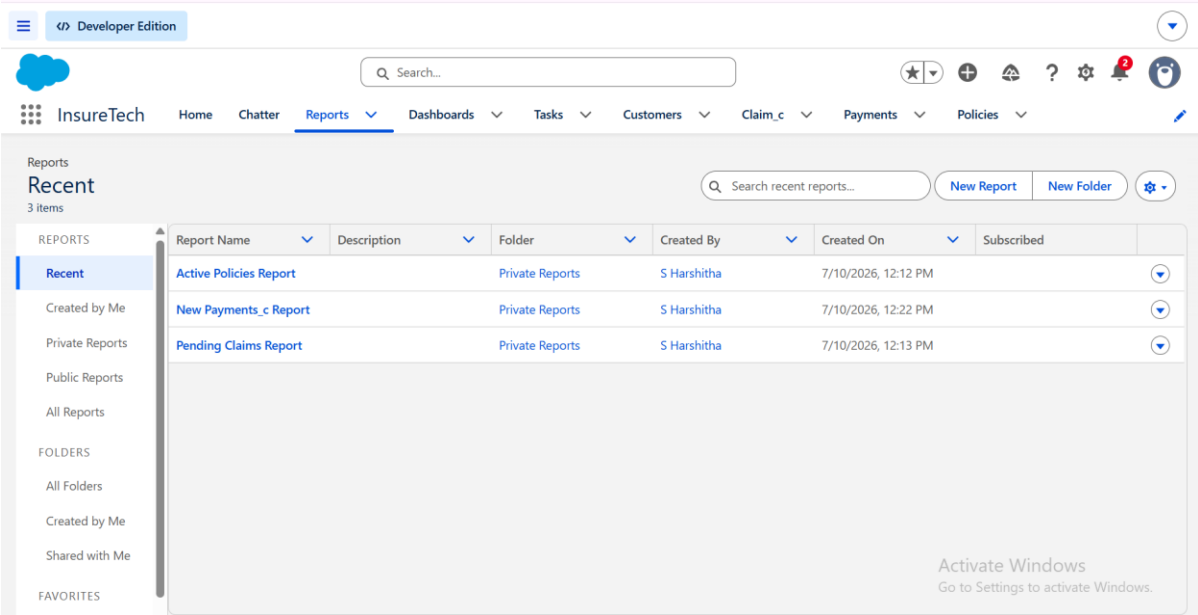
The Apex Trigger executes before inserting a Policy record and generates unique Policy Numbers such as POL-1001, POL-1002, and POL-1003. A handler class can be used to keep business logic separate from the trigger.

Figure: 11.6 Apex Policy Number Generation Workflow
(Insert Workflow Screenshot Here)

11.6 Reports and Dashboard Workflow

Operational data from Customers, Policies, and Claims is transformed into meaningful insights using Salesforce Reports and Dashboards.

Managers can monitor Total Customers, Total Policies, Total Claims, Active Policies, Claims by Status, and Policies by Type through interactive dashboard components.



11.7 Security Workflow

The Insurance Management System implements role-based security using Salesforce Profiles and Permission Sets. Insurance Agents, Claim Officers, Policy Managers, and Administrators are assigned appropriate permissions to access only the records and features required for their responsibilities. This approach protects sensitive customer information, prevents unauthorized modifications, and improves overall data security.

Figure: 11.8 Security Workflow
(Insert Security Workflow Screenshot Here)

12. Phase 4 – Data, Testing and Security

Sample records were created in Salesforce to verify customer registration, policy creation, claim processing, validation rules, Apex automation, Record-Triggered Flows, reports, dashboards, and security. Different user roles were tested using Permission Sets to ensure proper access control.

13. Test Cases

ID	Feature	Action	Expected Result	Status
TC-01	Premium Validation	Enter Premium ≤ 0	Record blocked	Pass
TC-02	Customer Creation	Create Customer	Customer created	Pass
TC-03	Policy Creation	Create Policy	Policy created	Pass
TC-04	Apex Trigger	Save Policy	Policy Number generated	Pass
TC-05	Policy Flow	Leave Status blank	Status = Active	Pass
TC-06	Claim Creation	Create Claim	Claim created	Pass
TC-07	Claim Flow	Approve Claim	Policy Status updated	Pass
TC-08	Reports	Open report	Relevant data displayed	Pass
TC-09	Dashboard	Refresh dashboard	Latest analytics displayed	Pass
TC-10	Relationships	Link records	Lookup works	Pass

14. Challenges and Resolutions

Challenge	Resolution
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Duplicate Policy Numbers	Generated automatically using Apex Trigger.
Invalid Premium	Validation Rule blocks invalid values.
Policy Status Update	Record-Triggered Flow updates status after claim approval.
Lookup Relationship Errors	Configured correct Customer–Policy–Claim relationships.
Dashboard Grouping	Used supported report groupings and charts.

15. Phase 5 – Deployment, Documentation and Maintenance

The project was developed and tested in a Salesforce Developer Org. Deployment can be performed using Change Sets or Salesforce CLI. Before deployment, validate custom objects, fields, relationships, validation rules, flows, Apex classes, reports, dashboards, Lightning pages, and permission sets.

15.1 Maintenance

- Monitor failed Flow interviews.
- Review Apex Debug Logs.
- Run Apex tests after code changes.
- Maintain picklist values.
- Review user permissions periodically.
- Validate reports after data model changes.
- Test new Flow versions before activation.
- Refresh dashboards regularly.

16. Future Enhancements

- AI-based claim fraud detection.
- Agentforce virtual assistant.
- Email and SMS policy renewal notifications.
- Experience Cloud customer portal.
- Payment gateway integration.
- Document upload and verification.
- Predictive analytics for risk assessment.
- Mobile application for customers and agents.

17. Conclusion

The Insurance Management System demonstrates the use of Salesforce Custom Objects, Lookup Relationships, Validation Rules, Formula Fields, Record-Triggered Flows, Apex

Triggers, Reports, Dashboards, and Lightning Apps to automate customer, policy, and claim management. The solution reduces manual effort, improves data accuracy, and provides a scalable foundation for modern insurance organizations.